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Satellite-based estimation of daily suspended sediment load using hybrid intelligent models

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ABSTRACT

This study uses a combination of support vector regression models, particle swarm optimization, and grey wolf optimization algorithms to predict suspended sediment load. For this purpose, The Satellite Precipitation of Climate Hazards Group InfraRed Precipitation with Station (CHIRPS) and Global Land Data Assimilation System (GLDAS) soil moisture products are utilized as the predictors. The prediction models are evaluated based on various visual and quantitative indicators. The Taylor and radar diagrams confirm that the support vector regression-particle swarm optimization best agrees with the observed values. Moreover, the obtained quantitative indices show that the support vector regression-particle swarm optimization model offers better performance than other models used in the present study. The values of the best indices are: Pearson correlation coefficient of 0.997, relative root mean square error of 13.17, percentage bias of 4.05, and Nash-Sutcliffe efficiency of 0.995.

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1 Introduction

Suspended sediment load (SSL) in a watershed is an essential hydraulic and hydrological parameter. Indeed, a lack of information on this parameter can disturb the performance of hydraulic structures such as water conveyance projects, decrease the volume of a reservoir significantly, cause low water quality and health problems, change waterways' topography and produce aquatic environmental issues (Guo *et al.* 2020, Ren *et al.* 2020).

SSL monitoring requires the installation of special financial and manpower-costly apparatus in hydrometric stations, which are in some cases impossible to instal in remote areas. Thus, alternative methods that can indirectly estimate the SSL at a lower cost are desirable in dealing with water resource management issues (Asadi *et al.* 2021). Conceptual, physical models have been developed to estimate SSL. The main obstacle for the conceptual and physical models is that they require high-resolution and high-quality information on flow and sediment, which is unavailable in most cases (Nourani *et al.* 2021). Another approach for SSL prediction is empirical models. Among these approaches, the most popular are sediment-rating curves (Horowitz 2003). However, high-quality experimental data is essential for this approach (Vercruysse *et al.* 2017).

Machine learning (ML) models can be used as an alternative for SSL estimation and other water resource management issues. ML employs algorithms to learn insights automatically and to recognize patterns from data and use this learning for better decision making. By studying and testing ML, programmers test the limitations that improve a computer system's perception, recognition, and performance. Support vector regression (SVR) is one of the most popular ML algorithms (Awad and Khanna 2015).

ML models have the advantages of simplicity, low computation time, high accuracy, and less information needed to predict hydrological phenomena (Yaseen *et al.* 2015). Artificial neural network (ANN) techniques, fuzzy-based models, and SVR are widely used ML models. In recent years, these features have made these models very popular in water resources management problems such as predicting water pollution concentration (Allah Nadiri *et al.* 2013), assessment of groundwater vulnerability (Baghapour *et al.* 2016, Nadiri *et al.* 2017, Hamamin and Nadiri 2018, Gharekhani *et al.* 2022), predicting groundwater levels (Nadiri *et al.* 2019, Khatibi and Nadiri 2021), wastewater treatment problems (Nadiri *et al.* 2018), and predicting SSL (Mianaei and Keshavarzi 2009, Rajaee 2011, Kisi 2012, Singh *et al.* 2013, He *et al.* 2014, Malik *et al.* 2019).

In recent years, in addition to applying these models solely to increase model performance, the combination of different algorithms, known as a hybrid model, has also been employed in SSL prediction. Banadkooki *et al.* (2020) used ANN-ant lion optimization (ALO), ANN-bat algorithm (BA), and ANN-particle swarm optimization (PSO) to predict the SSL. They noted that ANN hybrid models have a high aptitude to predict SSL (Banadkooki *et al.* 2020). To estimate SSL in the Mississippi River, Shadkani *et al.* (2021) used three learning models: multilayer perceptron (MLP), multi-layer perceptron-stochastic gradient descent (MLP-SGD), and gradient-boosted tree (GBT).

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Their comparative results showed the high accuracy of the hybrid MLP-SGD model compared to other models (Shadkani *et al.* 2021). Another series of hybrid models are based on fuzzy logic. These models can be trained faster than others and are more adaptive than individual ANNs or fuzzy logic (Kumar *et al.* 2019, Samet *et al.* 2019, Babanezhad *et al.* 2020). As a type of ML model, SVR is a new model that can be used instead of ANN. The SVR uses operational risk as a function instead of minimizing the error (Nourani *et al.* 2016).

Rezaei *et al.* (2021) compared the prediction ability of ANN, adaptive neuro-fuzzy inference systems (ANFIS), group method of data handling (GMDH), and least square support vector machines (LS-SVM) for estimation of the monthly SSL load in the Rudak station on the Jadjarood River, Iran. The results showed that the LS-SVM model has led to better results than other models from the perspective of root mean square error (RMSE), representing the model's capability of predicting the SSL load values.

Despite such an advantage, the SVR calculates its parameters based on trial and error, strongly affecting SSL prediction (Doroudi *et al.* 2021). Panahi *et al.* (2021) used the black widow optimization algorithm (BWOA) to develop two hybrid models, SVM-BWOA and ANFIS-BWOA. Their results indicated an increase in accuracy with hybrid models (Panahi *et al.* 2021).

In most SSL models, a reliable estimation is possible in the presence of an accurate spatio-temporal distribution of hydrological data such as precipitation, runoff, discharge, and soil moisture (SM), which is required for reliable SSL estimation (Himanshu *et al.* 2017). The spatio-temporal distribution of hydrological data such as precipitation and runoff discharge can be measured in a favourable situation by raingauge networks or hydrometric stations. In countries including Iran, the raingauge networks and hydrometric stations are usually unevenly distributed due to the complex and varying geographical and climatic conditions (Meng *et al.* 2014).

In SSL estimation, SM content plays a vital role. The adhesion and capillary forces that bind wet sediment particles increase with increasing SM. However, in the presence of high SM, favourable conditions are created for runoff, increasing erosion and sediment (Seeger *et al.* 2004). It is also challenging to measure SM at the watershed basin scale due to the large spatial and temporal variability and numerous SM-related factors (Korres *et al.* 2015). Satellite products can overcome some shortcomings (Ma *et al.* 2019). However, studies of the satellite and reanalysis products and sediment modelling are limited (Himanshu *et al.* 2017, Shrestha and Wang 2018, Ma *et al.* 2019, Fortesa *et al.* 2020). This means there is a dearth of studies using precipitation products and satellite SM in estimating SSL.

The resolution of satellite products is in the range of 0.05 to 0.25 degrees on different time scales (e.g. daily and monthly scales). Previous studies on the evaluation of satellite precipitation products have been performed based on ground observation data (Sorooshian *et al.* 2000, AghaKouchak *et al.* 2011, Gao and Liu 2013, Li *et al.* 2013, Jiang *et al.* 2016, Beck *et al.* 2017a). The World Meteorological Organization (WMO) maintains that more than 30 years of historical data are required for climate studies. The Climate Hazards Group InfraRed Precipitation with Station (CHIRPS) with high resolution (0.05 degree) satellite products provides 40 years of rainfall data (Funk *et al.* 2015). For streamflow forecasting, Sulugodu and Deka 92019) used CHIRPS satellite precipitation products and precipitation data from the Indian Meteorological Department (IMD). The results revealed the better performance of CHIRPS rainfall compared to IMD data. In addition, in the study of Beck *et al.* (2017b), 22 precipitation datasets were evaluated against the CHIRPS satellite precipitation products and a good performance was observed for the latter. Ghozat *et al.* (2020) studied the outcome of CHIRPS satellite precipitation from 1987 to 2017 in different climatic regions of Iran. The accuracy of the satellite products was compared with data from the 68 meteorological stations on different time scales (daily, monthly, and yearly) for different rainfall classes. Results showed that CHIPS performs well in a variety of climate zones, including mountainous regions, desert regions, and coastal regions. This is an essential advantage of the CHIRPS products in hydrological and climatic modelling.

Several studies have evaluated SM products globally (Brocca *et al.* 2011, Kim *et al.* 2015, Bai and Liu 2018, Ma *et al.* 2019). However, the validation process is still challenging and depends on numerous variables (Tavakol *et al.* 2019, Zhang *et al.* 2019). Saeedi *et al.* (2021) studied the performance of SM satellite products, including Soil Moisture Active Passive (SMAP), Advanced Microwave Scanning Radiometer (AMSR2), Global Land Data Assimilation System (GLDAS), and Catchment Land Surface Model (CLSM), against in situ observations in the Urmia Lake basin, Iran. The results showed the excellent performance of AMSR2 and GLDAS products.

Another challenging issue in SSL estimation using ML models is that the data from many sources can be included. However, computational problems in terms of sufficient memory and computer storage space increase the algorithm's limitations and complexity. Because some input data play a small role in forecasting and increase the required space for input data accompanied by the high complexity of the calculations, without reducing attention needed for prediction accuracy, attribute reduction techniques can reduce the input data space (Taghizadeh-Mehrjardi *et al.* 2016).

Principal component analysis (PCA) is used to identify the linear dependencies of variables and substitutes groups of the correlated variables with uncorrelated variables. It also reduces the dimensions of the input variables (Martinez and Kak 2001). Darabi *et al.* (2021) examined various parameters such as precipitation and discharge as input for three models (MLP, ANFIS, and Radial Basis Neural Network – RBFNN). The results showed that the integration of the optimization algorithms and the selection of input combinations for models based on the PCA technique improved the ability of the models to predict SSL. Also, Asadi *et al.* (2021) employed geomorphometric parameters and river discharge data as the main predictors in the sediment modelling process to estimate sediment load more accurately based on PCA in several river basins of Iran (Asadi *et al.* 2021).

In the present study, standalone and hybrid ML models (hybridized with optimization algorithms) are used to predict SSL in the watershed of a dam in southwestern Iran (i.e. Kowsar Dam), where sediment data is rarely available and

such a model could be beneficial. To this end, CHIRPS precipitation products and GLDAS SM have been selected as input predictors of the model. Two SVR models, SVR-PSO and SVR-grey wolf optimization (GWO), were also developed. Due to the large volume of satellite products, PCA was used to reduce the data dimension. In the end, the results of the ML models are compared with those of the sediment rating curve method. One innovation of this research is to employ satellite products of SM as one of the major factors in predicting SSL. More precisely, the objectives of this research are: (i) to develop an SSL prediction model based on the satellite products of SM and precipitation as the main predictors, and (ii) to assess the developed prediction model in places where sediment data are rarely available.

2 The case study

The Kowsar Dam watershed in Kohgiluyeh and Boyer-Ahmad Province, located in southwest Iran, was selected for study. This watershed is 2428 km^2 in area. Its mean annual rainfall is 687.8 mm. The Kosar reservoir dam was constructed in Tang-e Duk over the Kheirabad River. The average annual discharge of the Kheirabad river at the dam site is 22.65 m^3/s. The catchment elevation is in the range of 554 to 3487 m.a.s.l, and the average slope of the basin is 26.9%. Its annual sediment volume is 2.494 million tons. The approximate watershed longitude is from 50°26'42.2" to 51°15'36.6"E, and the latitude is between 30°27'0.7" and 30°55'53.1"N.

3 Data collection

To predict the SSL, the daily variables, including products of CHIRPS satellite precipitation and GLDAS, have been employed. Furthermore, the sediment simulation has been carried out using the daily observed sediment with geographical coordinates of 30°35'45.7"N and 50°37'30.8"E. Concerning the spatial resolution, the CHIRPS precipitation product (0.05° × 0.05°, 5.4 km) in the watershed has been gridded, and, consequently, the dam watershed grid is split into 216 different points. The precipitation products at the daily scale for each point have been extracted from 1986 to 2012. The CHIRPS data can be obtained from the web address https://www.chc.ucsb.edu.

To obtain the SM product, the watershed is gridded based on a spatial resolution of 0.25 × 0.25. To this end, the watershed is divided into 20 points. The SM products at the daily scale have been extracted for each point for the years 1986 to 2012. The GLDAS data are available at https://ldas.gsfc.nasa.gov/gldas. The location of Kowsar reservoir dam and the watershed gridding is shown in the satellite in Fig. 1.

4 Methods

To predict the SSL, the SVR method has been employed. Statistical specifications of satellite products and sediment suspended load are presented in Table 1. To present the predictive models, first, all satellite products of the rainfall and SM are re-scaled from x ϵ [a b] to x' ϵ [0 1], and their dimensions are normalized using the following equation:

$$\acute{x} = \frac{x - a}{b - a} \tag{1}$$

The data dimensions are decreased using the PCA technique. The PCA method selects normalized data of precipitation and satellite SM products as input vectors. The extracted vectors from the PCA method are then used as input variables for the predictive model.

In the training phase of the hybrid model, unknown parameters of the SVR method are considered the decision variables. To find the best quantities for the decision variables, GWO optimization algorithms and PSO are employed. Then, the SVR estimates the SSL values. These steps are iterated until the stopping criteria are satisfied. The developed models estimate the SSL based on the best solution in the testing phase. Finally, the performance of the prediction models is evaluated based on the four indices.

More details on the data dimension reduction, features selection, SVR method, optimization algorithms, and hybrid model are explained in the following sections. To better describe the present study methodology, a step-by-step flowchart of the proposed methodology is shown in Fig. 2.

4.1 Principal component analysis

While large datasets mostly include many variables, it might be possible to reduce the number of variables and maintain a large amount of essential information in the main dataset (Jolliffe 2005). Herein, the correlation between observational SSL on the origin day t and input precipitation data at 216 points, along with satellite SM data at 20 points of the watershed, are calculated, and the most appropriate data are identified. For example, the correlation between observational sediment of rainfall and satellite SM products at two points of the watershed (one point for networking of rainfall products and the other one based on networking of SM) is presented in Table 2.

Based on Table 2, the input variables, SM, and precipitation products are from the origin day (t) to 6 days earlier (t – 6) (the correlation ≥ 0.35). Thus, for each point, seven input variables, a total of $7 \times (216 + 20) = 1652$ predictors are used as the input, making the computation complex and slow. There are many techniques for decreasing the dimensions, and PCA is one of the most applicable methods. In this technique, the xy coordinate system can be substituted with a new orthogonal system, where the first axis passes the long data scatter. This framework prioritizes the first axis, which describes more variance, but the second axis helps a little. Due to this, the dimension data would decrease by dropping the second axis without losing much information as there is a significant correlation (Araghinejad 2014). Suppose there are p vectors of correlated vectors X_i where index i represents the ith vector. The principal components would be the linear combination of p vectors based on various parameters for each component. For the p vector, the p principal component can be achieved. The value of each principal component in the whole dataset is different. The p principal components are calculated as follows (Martinez and Kak 2001):

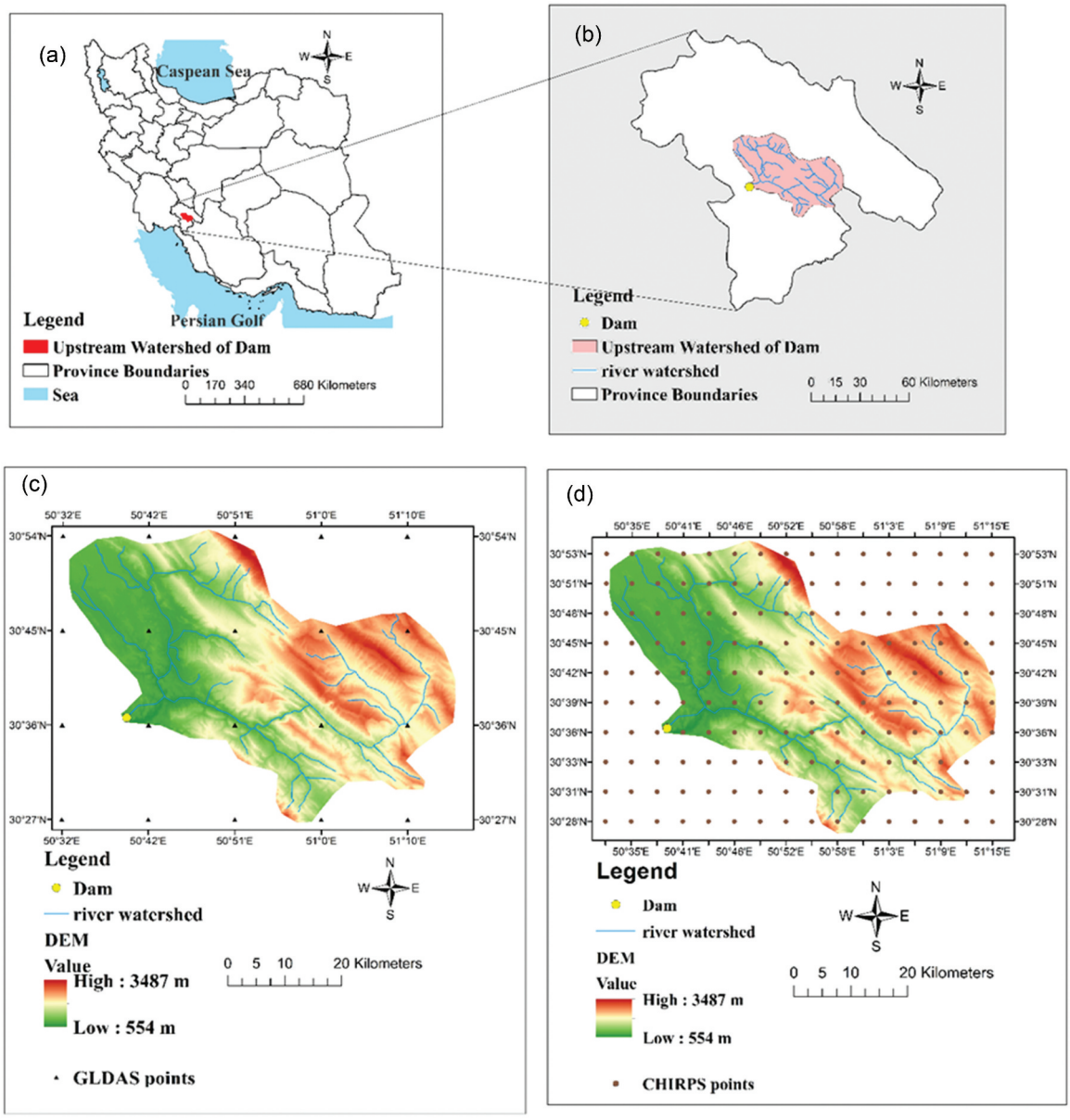


Figure 1. (a) Kohgiluyeh and Boyer-Ahmad Province and Kosar dam watershed in Iran; (b) Kosar dam watershed in higher scale; (c) Digital elevation model (DEM) of Kosar dam watershed with GLDAS points; (d) DEM of Kosar dam watershed with CHIRPS points.

Table 1. Statistics specifications of the observed sediment suspended load (SSL), precipitation and soil moisture data.

Parameters	Min	Max	Mean	SD	CV	Skewness
SSL(mg/L)	14.33	5497.00	340.94	889.88	2.61	4.29
P(mm)	8.00	5.39	5.70	1.06	105.99	0.00
SM	0.03	0.73	0.34	0.14	0.03	-0.11

SD: standard deviation, **CV**: coefficient variation, P: precipitation satellite production, SM: soil moisture satellite production

$$\begin{aligned} PC_1 &= Y_1 = a_{11}X_1 + a_{12}X_2 + \ldots a_{1p}X_p \\ PC_2 &= Y_2 = a_{21}X_1 + a_{22}X_2 + \ldots a_{2p}X_p \\ &\vdots \\ PC_p &= Y_p = a_{p1}X_1 + a_{p2}X_2 + \ldots a_{pp}X_p \end{aligned} \quad (2)$$

The first principal component includes the highest variance or eigenvalues, and the second principal component contains the second highest variance, and so on. The summation of all the PCs is equivalent to the total variance of the dataset. The variance is concentrated in the first few PCs that explain most of the dataset information.

4.2 Support vector regression

SVR is an approximation technique with good performance based on statistical learning theory. An SVR method is a regression framework that can provide a non-linear relation between various inputs and the target output. The main equation in this method can be shown as follows (Vapnik 2000):

$$f(x) = W^T\varphi(x) + b \quad (3)$$

where $f(x)$ indicates the function between target and input variables, W^T is the weighting vector of the input data, $\varphi(.)$ is

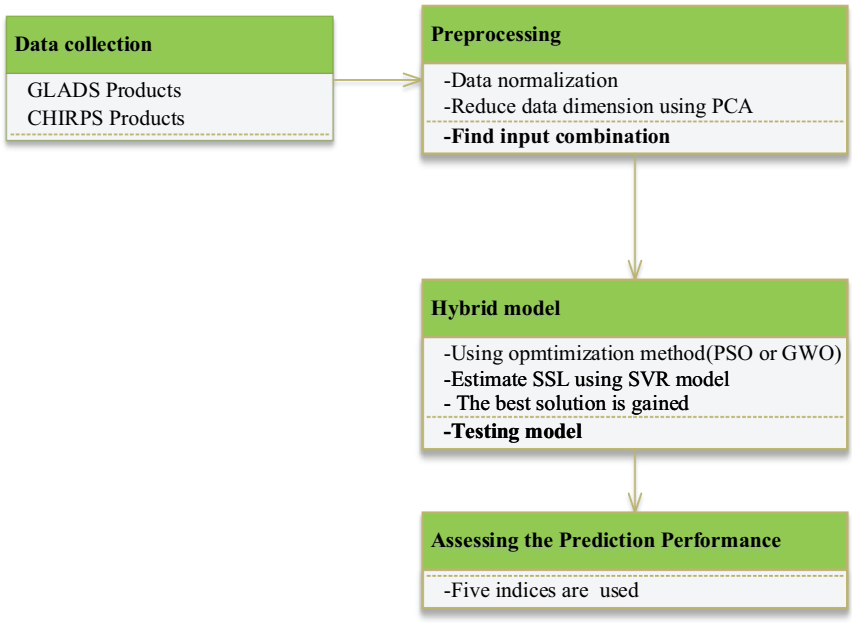


Figure 2. Step-by-step flowchart of the methodology proposed in this study.

Table 2. The correlation between the input variables (e.g., precipitation and soil moisture) and sediment suspended load (SSL).

precipitation	Correlation coefficient	soil moisture	Correlation coefficient
P (t)	0.58	SM (t)	0.39
P (t-1)	0.47	SM (t-1)	0.39
P (t-2)	0.44	SM (t-2)	0.39
P (t-3)	0.41	SM (t-3)	0.39
P (t-4)	0.36	SM (t-4)	0.35
P (t-5)	0.36	SM (t-5)	0.35
P (t-6)	0.35	SM (t-6)	0.35
P (t-7)	0.13	SM (t-7)	0.21
P (t-8)	0.12	SM (t-8)	0.21
P (t-9)	0.14	SM (t-9)	0.21
P (t-10)	0.10	SM (t-10)	0.01
P (t-11)	-0.03	SM (t-11)	0.01
P (t-12)	0.04	SM (t-12)	0.08
P (t-13)	0.08	SM(t-13)	0.01

Note: P refers to precipitation products, and SM refers to soil moisture products.

the non-linear transformation function that maps x into the m-dimensional feature vector, and b is the bias term.

The weights and bias are determined as the following equation is minimized (Vapnik 2000):

$$\begin{aligned} &Minimize: \tfrac{1}{2}||\mathrm{W}||^2 \textstyle\sum_{i=1}^{m}\left(\xi_i^- + \xi_i^+\right) \\ &subject\ to\ \left(w_i.\varphi(x_i) + b_{y_i}\right) - y_i \le \varepsilon + \xi_i^+ \quad i = 1.2....m \\ &y_i - \left(w_i.\varphi(x_i) + b_{y_i}\right) \le \varepsilon + \xi_i^- \quad i = 1.2....m \\ &\xi_i^-.\xi_i^+ \ge 0 \qquad i = 1.2....m \end{aligned} \tag{4}$$

where C is the penalty, ξ_i^+ and ξ_i^- are slack variables, W is the vector weight, m is the number of inputs, x is the input variables, y_i is the observational target variable, and ε is the insensitive loss function.

Using Lagrange multipliers, $\alpha_i.\alpha_i^-$ and the Karush–Kuhn–Tucker (KKT) condition Equation (3) modifies to (Vapnik 2000, Shawe-Taylor and Sun 2011):

$$f(x) = \sum_{i=1}^{m}\left(\alpha_i - \alpha_i^-\right)k\left(x_i.x_j\right) + b \tag{5}$$

where $k\left(x_i.x_j\right)$ is the kernel function. Several kernel functions, such as linear, sigmoid, polynomial, and radial basis function (RBF), can be used in the SVR models (Raghavendra and Deka 2014). In this study, RBF has been employed for sediment modelling using the following equation (Vapnik 2000):

$$k\left(x_i.x_j\right) = \exp\left(\frac{-\parallel x - x_i \parallel}{2\sigma^2}\right) \tag{6}$$

where σ is the kernel parameter.

Therefore, the SVR method has three unknown parameters: fine C, epsilon (ε), and σ. In this case, the best values can be determined using optimization algorithms (Rahgoshay *et al.* 2019, Doroudi *et al.* 2021).

It must be noted that an objective function must be determined in the optimization algorithm, which includes the RMSE function between observational and calculated sediment values in the case of this research. The RMSE of the SVR model is also calculated for comparison with the SVR-GWO and SVR-PSO models. The RMSE equation is as follows (Willmott 1981):

$$MSE = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(X_i - O_i\right)^2} \tag{7}$$

where the O_i and X_i are the observed SSL and predicted SSL, and m is the number of observed data.

4.3 Optimization methods

The GWO and PSO methods have enhanced the SVR method's performance. The GWO algorithm mimics the leadership hierarchy and hunting mechanism of grey wolves in nature. Four types of grey wolves – alpha, beta, delta, and omega – simulate the leadership hierarchy. In addition, three main stages of hunting – (i) searching for prey, (ii) encircling prey, and (iii) attacking prey – are followed. Concerning these three stages, search agents are required to change their positions, which ensures the convergence of an algorithm in the search space. (Mirjalili *et al.* 2014). Once the agent set is determined, each agent's position in each iteration is updated using Equations (8–10):

$$\vec{D}_\alpha = \left|\vec{C}_1 \cdot \vec{X}_\alpha(t) - \vec{X}(t)\right|.\vec{D}_\beta = \left|\vec{C}_2 \cdot \vec{X}_\beta(t) - \overrightarrow{X(t)}\right|.$$
$$\vec{D}_\delta = \left|\vec{C}_3 \cdot \vec{X}_\delta(t) - \overrightarrow{X(t)}\right| \quad (8)$$

$$\vec{X}_1(t) = \vec{X}_\alpha(t) - \vec{A}_1 \cdot \left(\vec{D}_\alpha\right).\vec{X}_2(t) = \vec{X}_\beta(t) - \vec{A}_2 \cdot \left(\vec{D}_\beta\right).$$
$$\vec{X}_3(t) = \vec{X}_\delta(t) - \vec{A}_3 \cdot \left(\vec{D}_\delta\right) \quad (9)$$

$$\vec{X}_{(t+1)} = \frac{\vec{X}_1(t) + \vec{X}_2(t) + \vec{X}_3(t)}{3} \quad (10)$$

where t represents the current iteration, $\vec{A}$ and $\vec{C}$ are coefficient vectors, $\vec{X}_{\alpha.\beta.\delta}$ are the position vectors of the prey, and $\vec{X}_{1.2.3}$ is the position vector of the grey wolf.

The choreography of a flock of birds inspired the PSO algorithm. In this approach, the behaviour of each individual is influenced by either the best local or the best global individual. This approach uses population and performance measurement concepts similar to the fitness value used with evolutionary algorithms. PSO also allows individuals to use their past experiences so search agents can search more broadly in the search space (Eberhart *et al.* 2001). Based on the following relations, the location and velocity of the particle particles are updated:

$$V_i^{t+1} = \xi V_i^t + c_1 r_1 \left(x_{ibest}^t - x_i^t\right) + c_2 r_2 \left(x_{igbest}^t - x_i^t\right) \quad (11)$$

$$x_i^{t+1} = x_i^t + v_i^t \cdot t \quad (12)$$

where V_i^{t+1} is the velocity of the particle at iteration t + 1, ξ is the inertia weight, r_1 and r_2 are random numbers, x_{igbest}^t is the best group position, x_{igbest}^t and c_2 are acceleration coefficients, x_i^t is the position of particles at iteration t, and x_i^{t+1} is the position of the particle at iteration t + 1.

4.4 Description of hybrid models

In this article, the PSO or GWO optimization algorithm adjusts key SVR parameters with unknown values. RMSE was employed as an objective function of the hybrid models. This process determines input data, data mining, and input data combinations. The steps for development of the SVR-GWO model are as follows:

(i) The initial population of wolves is created based on a set of random solutions.
(ii) The target values for each wolf are predicted based on the SVR method, and the target function is calculated.
(iii) The first three wolves are selected and stored as alpha, beta, and delta.
(iv) The position of the wolves is updated using Equations (8–10), and the target values for each wolf are calculated based on the SVR.
(v) Alpha, beta, and delta wolves are updated.
(vi) The wolf alpha position and score are returned as the best solution.
(vii) The steps above are repeated to meet the termination criteria.

The SVR-GWO model algorithm is presented in Fig. 3.

The second hybrid model is the SVR-PSO method. The following steps are used to develop this model:

(i) The initial population of the PSO algorithm is determined together with the random parameters of the algorithm.
(ii) The SVR model predicts target values based on the training data. The value of the objective function of the model is calculated.
(iii) The velocity and position of the particles, which represent the optimal value of the three parameters of the backup vector method, are calculated using Equations (11) and (12).
(iv) The best solution, the best global individual, and the best local individual are updated.
(v) The steps above are repeated to meet the termination criteria.

The SVR-PSO model algorithm is shown in Fig. 4.

4.5 Evaluation of the predictive models

Four indicators are used to evaluate the predictive ability of the SVR, SVR-PSO, and SVR-GWO models. These indicators are Pearson correlation coefficient (CC, where a higher value of CC indicates a more accurate forecast), relative root mean squared error (RRMSE, where a lower value of RRMSE indicates higher accuracy of forecast) (Mouillot and Leprêtre 1999), percentage bias (PBIAS, where a lower value of PBIAS shows a more accurate forecast) (Moriasi *et al.* 2007), and Nash-Sutcliffe efficiency (NSE, where a higher value of NSE indicates greater accuracy in predicting) (Nash and Sutcliffe 1970). These indices are defined as follows:

$$\mathrm{CC} = \left[\frac{\sum_{i=1}^{m}(O_i - \bar{O})(X_i - \bar{X})}{\sqrt{\sum_{i=1}^{m}(O_i - \bar{O})^2 \sum_{i=1}^{m}(X_i - \bar{X})^2}}\right] \quad (13)$$

$$RRMSE = 100 \times \frac{RMSE}{\bar{O}} = 100 \times \frac{\sqrt{\frac{1}{m}\sum_{i=1}^{m}(X_i - O_i)^2}}{\bar{O}} \quad (14)$$

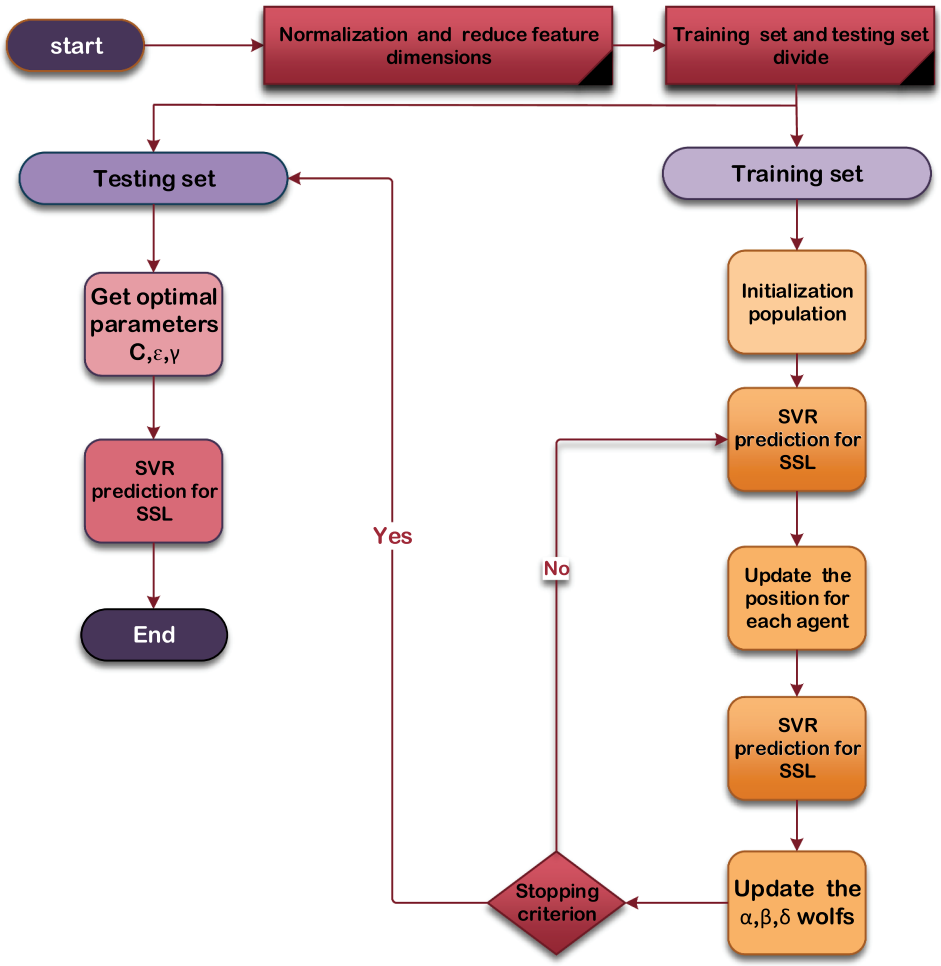


Figure 3. Flowchart of the SVR-GWO algorithm model used in this study.

$$NSE = 1.0 - \left[\frac{\sum_{i=1}^{m}(O_i - X_i)^2}{\sum_{i=1}^{m}(O_i - \bar{O})^2}\right] \quad (15)$$

$$PBIAS = 100 \times \left[\frac{\sum_{i=1}^{m}(O_i - X_i)}{\sum_{i=1}^{m} X_i}\right] \quad (16)$$

where O_i and X_i are observed and predicted SSL, respectively. $\overline{O}$ is the average of observed SSL, and m is the number of observed data.

4.6 Sediment rating curve

A rating curve involves a diagram or equation that can be used to estimate sediment loads. The power function is the most popular equation used as a rating curve (Walling 1977).

$$Q_s = aQ^b \quad (17)$$

in which Q_s is the daily mean suspended sediment discharge (in kg/s), Q is the daily mean water discharge (in m^3/s), and a and b are the rating curve parameters.

5 Results

Different training–testing divisions (73–27%, 72–28%, 71–29%, 70–30%, 69–31%, 68–32%, and 67–33%) are assessed to find the best division. The RRMSE and NSE values obtained from those divisions are presented in Fig. 5. Figure 5 shows that the 70–30% division offers the best performance for all models used in the current study. Hence, this division is used to assess the models for estimating the daily suspended sediment load.

5.1 Data mining and input feature selection

In this study, using Equation (1), precipitation and SM data are normalized. Consequently, due to the extensive dimensions of the input data, the input data are reduced, and the inputs (i.e. the features) are selected based on the PCA method. The results of the PCA are shown in Table 3.

Based on Table 3, the 10 main components, which include 91.73% of the eigenvalues of all the components, have been selected as the input of the models. Therefore, the input combinations are (M1, M2 ... M10). M10 includes all major components, M9 has nine components, and ... M1 consists of only the first major component.

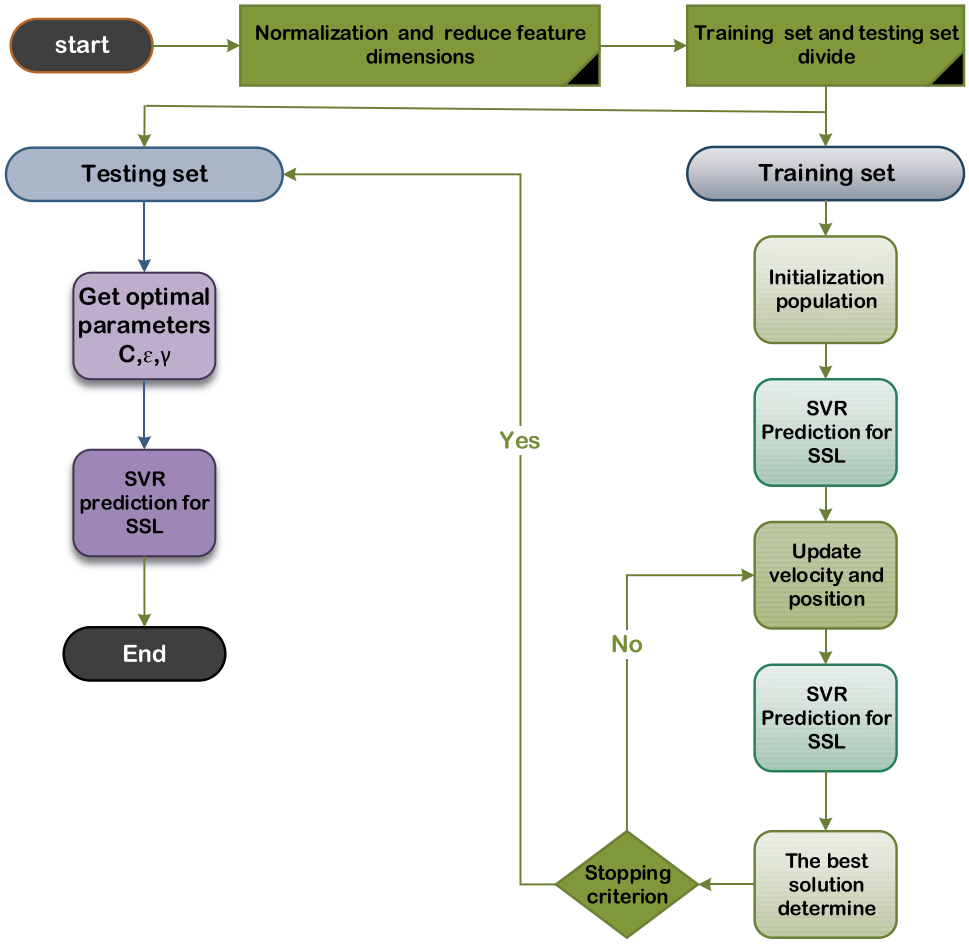


Figure 4. Flowchart of the SVR-PSO algorithm model used in this study.

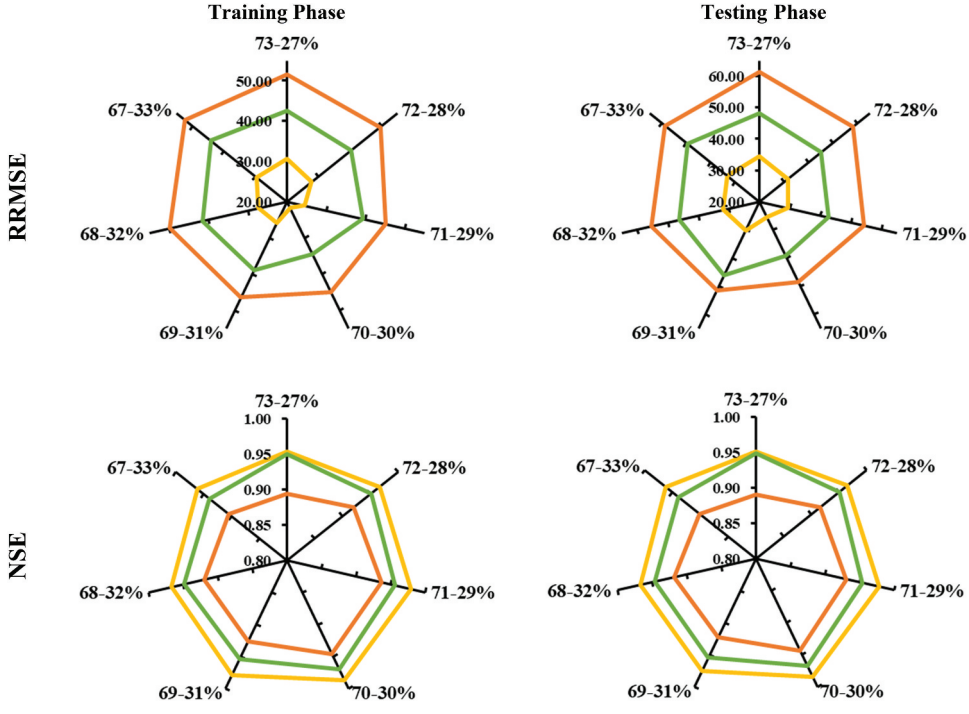


Figure 5. The values of the RRMSE and NSE indices obtained from different training–testing divisions.

Table 3. The results obtained by principal component analysis.

Principal component	Eigenvalues	Percentage
1	1.9319	43.66
2	0.7036	15.90
3	0.4819	10.89
4	0.2406	5.44
5	0.2039	4.61
6	0.2009	4.54
7	0.1071	2.42
8	0.0939	2.12
9	0.0537	1.21
10	0.0416	0.94
Last principal component	1.69E-32	3.81E-31

5.2 Evaluation of model performance

To compare the performance of the models, i.e. SVR, SVR-GWO, SVR-PSO, the metric indices for 10 different input combinations in the training phase and testing phase (using Equations 13–16) are shown in Tables 4–6, and the objective function of the models, which is the value of RMSE, is shown in Fig. 6.

Based on the results of Table 4 for the SVR model, the best values, which are CC = 0.99134, RRMSE = 29.85, PBIAS = 13.80516, and NSE = 0.96640, belong to the SVR-M7 model. In other words, out of four criteria, the top four criteria result from the SVR-M7 model. In the testing phase, the results show that the best values, which are CC = 0.99055, RRMSE = 38.42, PBIAS = 18.71480 and NSE = 0.96006, are obtained with the SVR-M7 model. Figure 6(a) also shows that the lowest value of the objective function (RMSE) in the training and testing phases is obtained in the SVR-M7 model. Hence, the best performance among the SVR models belongs to the SVR-M7 model.

As shown in Table 5, in the training phase, the highest values of CC = 0.99768 and NSE = 0.99635 are associated with the SVR-PSO-M7 model. As shown in Table 5, the lowest values of the criteria RRMSE = 13.17 and PBIAS = 4.03747 were obtained by the SVR-PSO-M7 model. Likewise, in the testing phase, the best performance is related to SVR-PSO-M7 (CC = 0.99700, RRMSE = 17.22, PBIAS = 4.05122, and NSE = 0.99499). In addition, Fig. 6(b) shows that in the training and testing phases, the lowest RMSE was obtained by the SVR-PSO-M7 model. Thus, this model has the best performance among the SVR-PSO models.

As far as the SVR-GWO model is concerned, based on the results of Table 6, in the training phase the SVR-GWO-M7 model has the best performance in terms of all criteria compared to other SVR-GWO models (CC = 0.99287, RRMSE = 24.22, PBIAS = 8.95907, NSE = 0.98104). In the testing phase, the highest values of CC = 0.99204 and NSE = 0.97847 were achieved by the SVR-GWO-M7 model. Likewise, the lowest values of RRMSE = 29.21 and PBIAS = 12.68903 are related to the same model. Also, Fig. 6(c) illustrates that the best value of RMSE, which is the objective function, is obtained with the SVR-GWO-M7 model. Therefore, the best performance among SVR-GWO hybrid models is SVR-GWO-M7.

The criteria achieved from the various prediction models (SVR, SVR-PSO, and SVR-GWO) confirm that the M7 input combination, which consists of seven main components, is the best for SSL prediction. The SVR-M7, SVR-PSO-M7, and SVR-GWO-M7 have been selected for further evaluation. The network structures of the best models are shown in Fig. 7.

To find the best-fit model among all the models introduced in the present study, a radar diagram (Fig. 8) has been used as a visual comparison tool. According to Fig. 8, the highest value

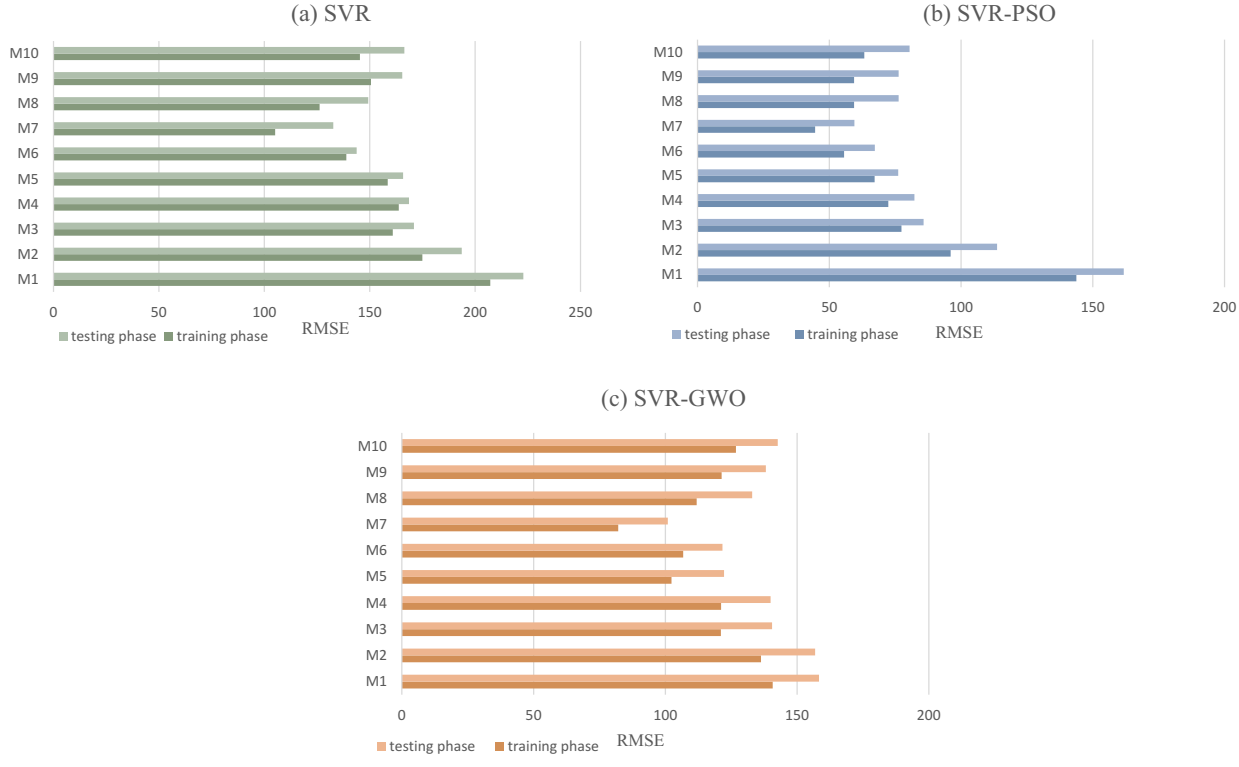


Figure 6. The RMSE values obtained by models (a) SVR; (b) SVR-PSO; (c) SVR-GWO.

Table 4. Values of the performance indicators obtained by the SVR model.

	Train			
Model	CC	RRMSE	NSE	PBIAS
SVR-M10	0.98346	42.89	0.94791	20.87372
SVR-M9	0.98371	44.43	0.94779	18.94186
SVR-M8	0.99030	37.24	0.95188	16.03856
SVR-M7	0.99134	29.85	0.96640	13.80516
SVR-M6	0.98992	40.98	0.95035	20.85711
SVR-M5	0.98927	46.78	0.94259	26.11811
SVR-M4	0.98843	52.67	0.94035	26.89636
SVR-M3	0.98598	44.53	0.94872	24.24709
SVR-M2	0.98250	51.63	0.93730	27.66267
SVR-M1	0.98107	58.18	0.92632	30.25932
	Test			
Model	CC	RRMSE	NSE	PBIAS
SVR-M10	0.98310	48.17	0.94764	24.01445
SVR-M9	0.98214	47.89	0.94701	23.85766
SVR-M8	0.98940	43.22	0.95076	19.32207
SVR-M7	0.99055	38.42	0.96006	18.71480
SVR-M6	0.98907	41.63	0.94958	23.02385
SVR-M5	0.98789	48.00	0.94486	26.51553
SVR -M4	0.98631	53.59	0.94028	26.22049
SVR-M3	0.98444	46.59	0.93966	26.84112
SVR-M2	0.97921	56.05	0.93454	30.16103
SVR-M1	0.97811	61.62	0.92269	32.45562

Table 5. Values of the performance indicators obtained by the SVR-PSO model.

	Train			
Model	CC	RRMSE	NSE	PBIAS
SVR-PSO-M10	0.99007	18.67	0.98968	7.45014
SVR-PSO-M9	0.99093	17.52	0.99031	6.56091
SVR-PSO-M8	0.99082	17.52	0.99021	6.53472
SVR-PSO-M7	0.99768	13.17	0.99635	4.03747
SVR-PSO-M6	0.99306	16.41	0.99089	5.96316
SVR-PSO-M5	0.99142	19.83	0.98980	6.47911
SVR-PSO-M4	0.99016	21.36	0.98803	7.63907
SVR-PSO-M3	0.99077	22.83	0.98704	8.12495
SVR-PSO-M2	0.98961	28.34	0.98174	12.60577
SVR-PSO-M1	0.98834	42.41	0.96054	22.34007
	Test			
Model	CC	RRMSE	NSE	PBIAS
SVR-PSO-M10	0.99020	23.29	0.98867	9.76575
SVR-PSO-M9	0.99029	22.09	0.98941	8.34117
SVR-PSO-M8	0.99029	22.08	0.98930	8.33659
SVR-PSO-M7	0.99700	17.22	0.99499	4.05122
SVR-PSO-M6	0.99255	19.48	0.98987	5.82366
SVR-PSO-M5	0.99120	22.04	0.98844	7.47284
SVR-PSO-M4	0.99009	23.83	0.98733	9.86389
SVR-PSO-M3	0.99005	24.83	0.97967	11.23271
SVR-PSO-M2	0.98736	32.90	0.98037	16.31902
SVR-SPO-M1	0.98433	46.81	0.96039	24.86982

of the criteria (CC, NSE) in the training and testing phases are related to the SVR-PSO-M7 model. Also, as can be seen, the best values of the RRMSE and PBIAS criteria in the training and testing phases belong to the SVR-PSO-M7 model. Therefore, the SVR-PSO-M7 has a better performance than the two other models.

To find the most appropriate of the responding patterns among the selected models, scatter plots between the estimated SSL and the measured SSL are plotted in Fig. 9. The scatter plots can show the degree of correlation – if there is any – between estimated values and observational data (Touchette *et al.* 1985).

Figure 9(a) illustrates that the value of R^2 (coefficient of determination) in the three selected models is equal to $R^2_{SVR-PSO-M7} = 0.995$. $R^2_{SVR-GWO-M7} = 0.986$. $R^2_{SVR} = 0.983$. Therefore, the SVR-PSO-M7 model performs slightly better in the training phase than other models. Also, Fig. 9(b) shows that the best performance belongs to the SVR-PSO-M7, which differs from the SVR-M7 model by $\frac{0.994-0.981}{0.981} \times 100 = 1.31\%$, and with the SVR-GWO-M7 model by $\frac{0.994-0984}{0.984} \times 100 = 1.01\%$. Therefore, the SVR-PSO-M7 performs slightly better than other models in the training and testing phases.

For a more accurate qualitative assessment of the performance of the models, the SSL observations are plotted against the estimated SSL in Fig. 10. Also, for quantitative assessment, different quartiles of the observed and estimated data are given in Table 7. As shown in Fig. 10, the PSO-M7 SVR model is rated better than other SSL models in the training and testing phases.

As shown in Table 7, in the training phase, the minimum and maximum SSL_{25} are, respectively, the observed data ($SSL_{25} = 66.67\ mg/L$) and SVR-M7 ($SSL_{25} = 157.43\ mg/L$). The closest SSL_{25} value to the observed data belongs to the model ($SSL_{25} = 81.63\ mg/L$) SVR-PSO-M7. The minimum and maximum SSL_{50} are the observed data (($SSL_{25} = 81.63\ mg/L$) and SVR-M7 ($SSL_{50} = 159.48\ mg/L$), respectively. The closest SSL_{50} value to the observational data belongs to model SVR-PSO-M7 ($SSL_{50} = 101.74\ mg/L$). The lowest value of SSL_{75} among the models belongs to the model SVR-PSO-M7 ($SSL_{75} = 157.38\ mg/L$), and the highest value belongs to the model SVR-M7 ($SSL_{75} = 157.38\ mg/L$. The relative difference in the data observed with the SVR-M7, SVR-PSO-M7, and SVR-GWO-M7 models is 24.29%, 0.61%, and 19.13%, respectively, so the closest value in 75% of the observational data is for the SVR-PSO-M7 model. Examining the results of the testing phase revealed the minimum and maximum 25% of the data in the

Table 6. Values of the performance indicators obtained by the SVR-GWO model.

	Train			
Model	CC	RRMSE	NSE	PBIAS
SVR-GWO-M10	0.98906	37.40	0.96464	20.37752
SVR-GWO-M9	0.98965	35.80	0.96943	19.35097
SVR-GWO-M8	0.99082	33.00	0.97337	15.26246
SVR-GWO-M7	0.99287	24.22	0.98104	8.95907
SVR-GWO-M6	0.99087	31.49	0.97485	13.59710
SVR-GWO-M5	0.99043	30.18	0.97609	14.20223
SVR-GWO-M4	0.98983	35.74	0.97049	17.96837
SVR-GWO-M3	0.98911	35.72	0.97052	16.81876
SVR-GWO-M2	0.98549	40.21	0.96531	19.60748
SVR-GWO-M1	0.98332	41.52	0.95368	21.68126
	Test			
Model	CC	RRMSE	NSE	PBIAS
SVR-GWO-M10	0.98900	41.29	0.96296	17.30022
SVR-GWO-M9	0.98860	39.98	0.96740	17.09087
SVR-GWO-M8	0.99023	38.48	0.96999	13.62791
SVR-GWO-M7	0.99204	29.21	0.97847	12.68903
SVR-GWO-M6	0.99065	35.21	0.97225	13.76125
SVR-GWO-M5	0.99015	35.39	0.97008	20.63135
SVR-GWO-M4	0.98941	40.50	0.96783	20.70740
SVR-GWO-M3	0.98627	40.66	0.96765	35.43753
SVR-GWO-M2	0.98596	45.40	0.96215	35.49752
SVR-GWO-M1	0.98302	45.82	0.95163	34.57928

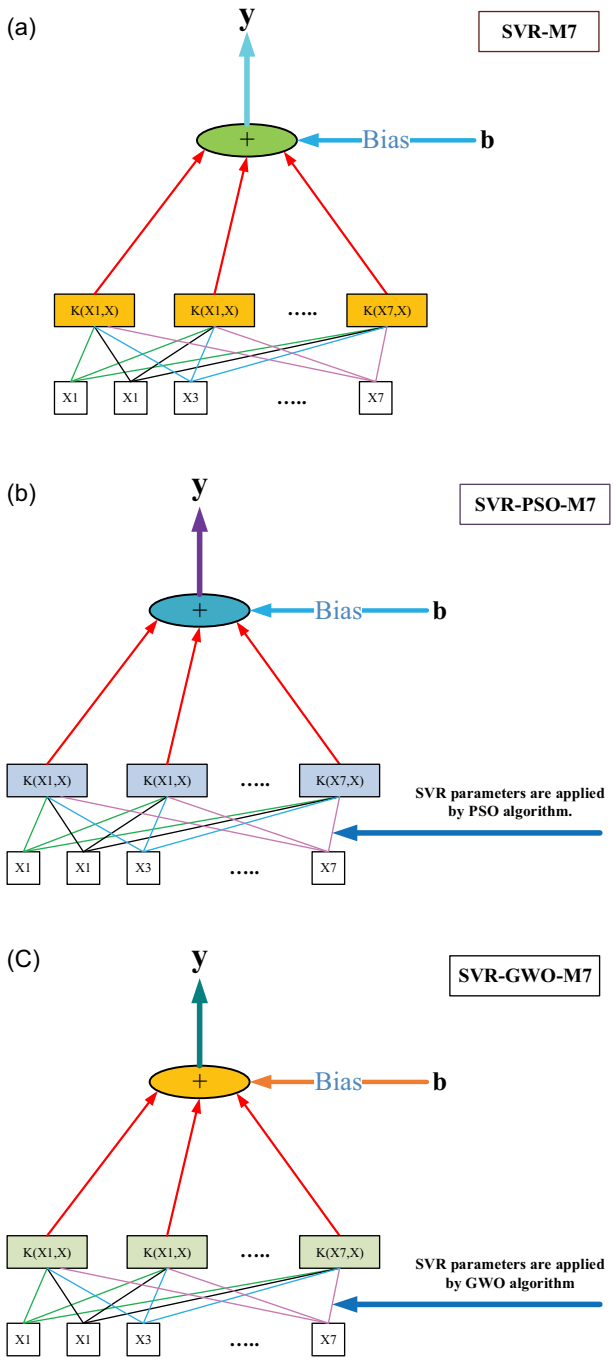


Figure 7. The network structures of the best models: (a) SVR-M7; (b) SVR-PSO-M7; (c) SVR-GWO-M7.

observed data ($SSL_{25} = 56.67\ mg/L$) and SVR-M7 ($SSL_{25} = 194.93\ mg/L$), respectively. The closest value of SSL_{25} to the observational data belongs to model SVR-PSO-M7 ($SSL_{25} = 77.34\ mg/L$). The minimum and maximum SSL_{50} belong to the observational data ($SSL_{50} = 97.75\ mg/L$) and SVR-M7 ($SSL_{50} = 199.03\ mg/L$), respectively. The relative differences between the observed data and SVR-M7, SVR-PSO-M7, and SVR-GWO-M7 are 103.61%, 3.36%, and 54.52%, respectively, so the closest value of SSL_{50} to the observational data belongs to model SVR-PSO-M7 ($SSL_{50} = 101.04\ mg/L$). The lowest value of SSL_{75} belongs to the model SVR–PSO-M7 (($SSL_{50} = 101.04\ mg/L$)), and the highest value belongs to the model SVR-M7 ($SSL_{75} = 228.79\ mg/L$). The relative difference of the observed data from the models SVR-M7, SVR-PSO-M7, and SVR-GWO-M7 is 63.42%, 2.03%, and 33.04%, respectively. So, the closest value in 75% of the observational data is the SVR-PSO-M7 model. Finally, it can be concluded that the lowest differences in the simulated data compared to the observational data in both training and testing phases belong to the SVR-PSO-M7 model.

To compare the qualitative agreement of prediction models with observed SSL in terms of correlation, RMSE, and standard deviation variance, Taylor diagrams (Taylor 2001) have been used. Figure 11 shows Taylor diagrams for the training and testing phases. As shown in Fig. 11(a), all three models are very close to the observed data in terms of correlation. However, the SVR-PSO-M7 model is slightly more consistent with the observed data than the other two models. Also, in terms of deviating from the standard deviation and RMSE criteria, the combined model SVR-PSO is closer to the observed data. Figure 11(b) shows that all three correlation models are closely related to the observed data. However, the SVR-PSO model is slightly more consistent with the observed data than the other models. In the case of RMSE and standard deviation, the SVR-PSO model produces superior performance over other models.

5.3 Comparison between results of sediment rating curve and SVR models

Since the sediment data in the studied region are inadequate, sediment data are classified into five sections to improve the estimation accuracy of the sediment rating curve (SRC) method. Each section contains rating parameter characteristics which are tabulated in Table 8.

The SSL is estimated employing the parameters obtained in Table 8, and the values of the criteria in the training phase are found to be CC = 0.97879, RRMSE = 52.42, PBIAS =4.17547, NSE = 0.95804. Compared to the selected SVR-PSO-M7 model and based on the results given in Table 5, the CC correlation criterion $\left(100 \times \frac{0.97879-0.99768}{0.97879}\right) = -1.93\%$ is less than the NSE criterion $\left(100 \times \frac{0.95804-0.99635}{0.95804}\right) = -4\%$, indicating the higher accuracy of the SVR-PSO-M7 model than the SRC method. In the case of the RRMSE and PBIAS criteria, the values of the SRC model are 298% and 3.3% higher than the selected model, respectively, suggesting a better estimation of the SVR-PSO-M7 model. Comparing the sediment rating curve method and the SVR-GWO-M7 model, the correlation criterion is equal to $\left(100 \times \frac{0.95804-0.99635}{0.95804}\right) = -4\%$ NSE -2.4%, 116.4% RRMSE, which indicates the better performance of the SVR-GWO-M7 model in these three criteria. However, in the case of the PBIAS criterion, the SRC method gives a better performance, according to the $\left(100 \times \frac{5.17547-8.95907}{5.17547}\right) = -73\%$ difference. Hence, it can be said that the SVR-GWO-M7 reveals an almost entirely better performance than SRC. According to Table 4, the comparison between results of the sediment rating curve method and SVR-M7 based on the CC, RRMSE, PBIAS, and NSE criteria is equal to −1.28%, 75.6%, −166%, and −0.88%, respectively, representing the almost entirely better performance of the SVR-M7 model (i.e. better except in the case of

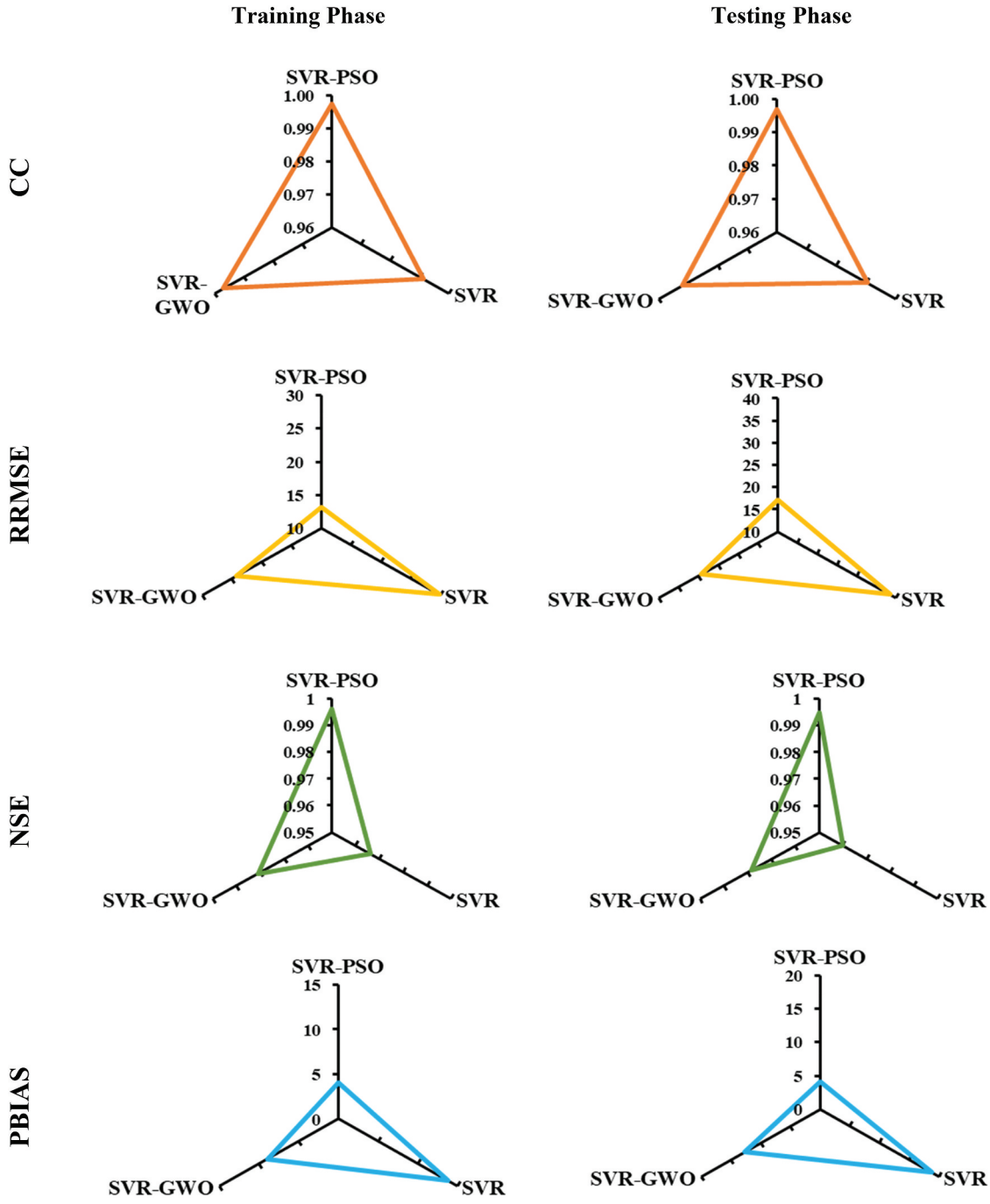


Figure 8. The performance of the selected models.

the PBIAS criterion). The values of the criteria in the testing phase are CC = 0.95385, RRMSE = 82.81, PBIAS = 9.43784, and NSE = 0.90734. According to Tables 4–6, the models SVR-M7 and SVR-GWO-M7 are better than the SRC method in terms of the CC, RRMSE, and NSE criteria, while the SVR-PSO-M7 performs better than SRC in terms of all four criteria.

6 Discussion

Evaluation of the results of SSL prediction shows that accurate determination of random and unknown parameters of the SVR method using optimization algorithms increases the prediction accuracy. Also, the data in Tables 4–6 show that the combined SVR-PSO and SVR-GWO models outperformed the SVR model on all four criteria. This is consistent with previous research observations (Rahgoshay *et al.* 2019, Panahi *et al.* 2021) on SVR hybrid models.

Doroudi *et al.* (2021) examined the watershed of Kowsar Dam and compiled support vector regression-genetic algorithm (SVR-GA) and support vector regression-observer teacher learner based optimization (SVR-OTLBO) to predict daily sediment suspended load, with CC = 0.9769 and RRMSE = 0.53716. Table 5 and Fig. 9 show that the correlation between the estimated and observed data (i.e. CC = 0.99768, R^2 = 0.995) is superior. Furthermore, RRMSE and RMSE are important and simply interpretable validation criteria as they

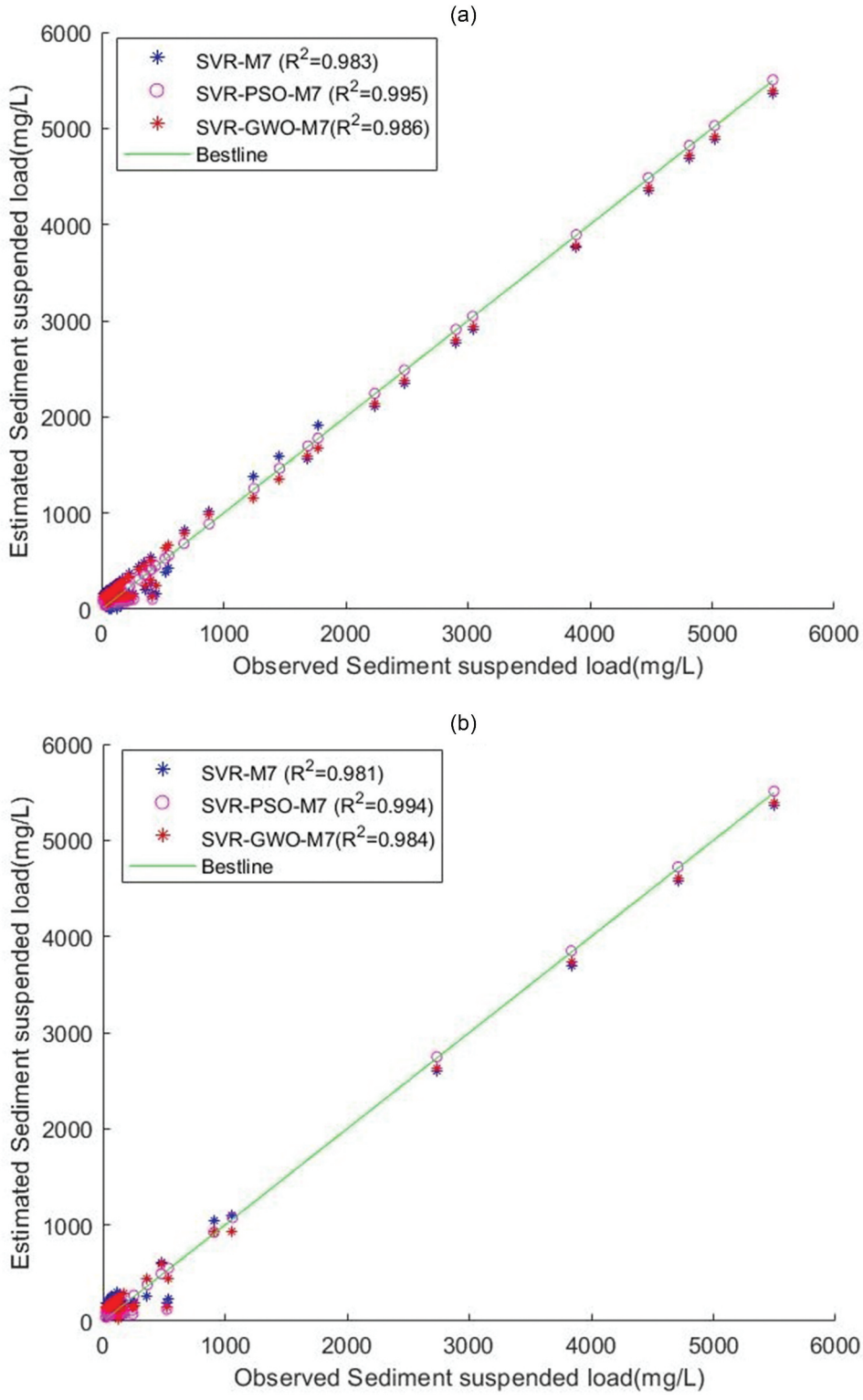


Figure 9. Scatter plots of observed and estimated suspended sediment load (mg/L): (a) training phase and (b) testing phase.

specify the actual size of the error produced by the model. This study shows that the best combined model (SVR-PSO-M7) obtained RRMSE = 13.17% = 0.1317. Therefore, the evaluation criteria for the models developed in this study are superior to the previous one (Doroudi *et al.* 2021). Although RRMSE has many advantages over CC and R^2, larger errors, when squared, exaggerate the effect of those errors on the sum of the square errors (Willmott *et al.* 2012). In this study, the NSE criterion was used to solve this problem. A value of NSE closer to 1 shows the better performance of the prediction models (Nash and Sutcliffe 1970). The results of the model validation indicate that the best value obtained is NSE = 0.99499. It could be acknowledged that using satellite products (precipitation and SM) in sediment modelling increases correlation and reduces

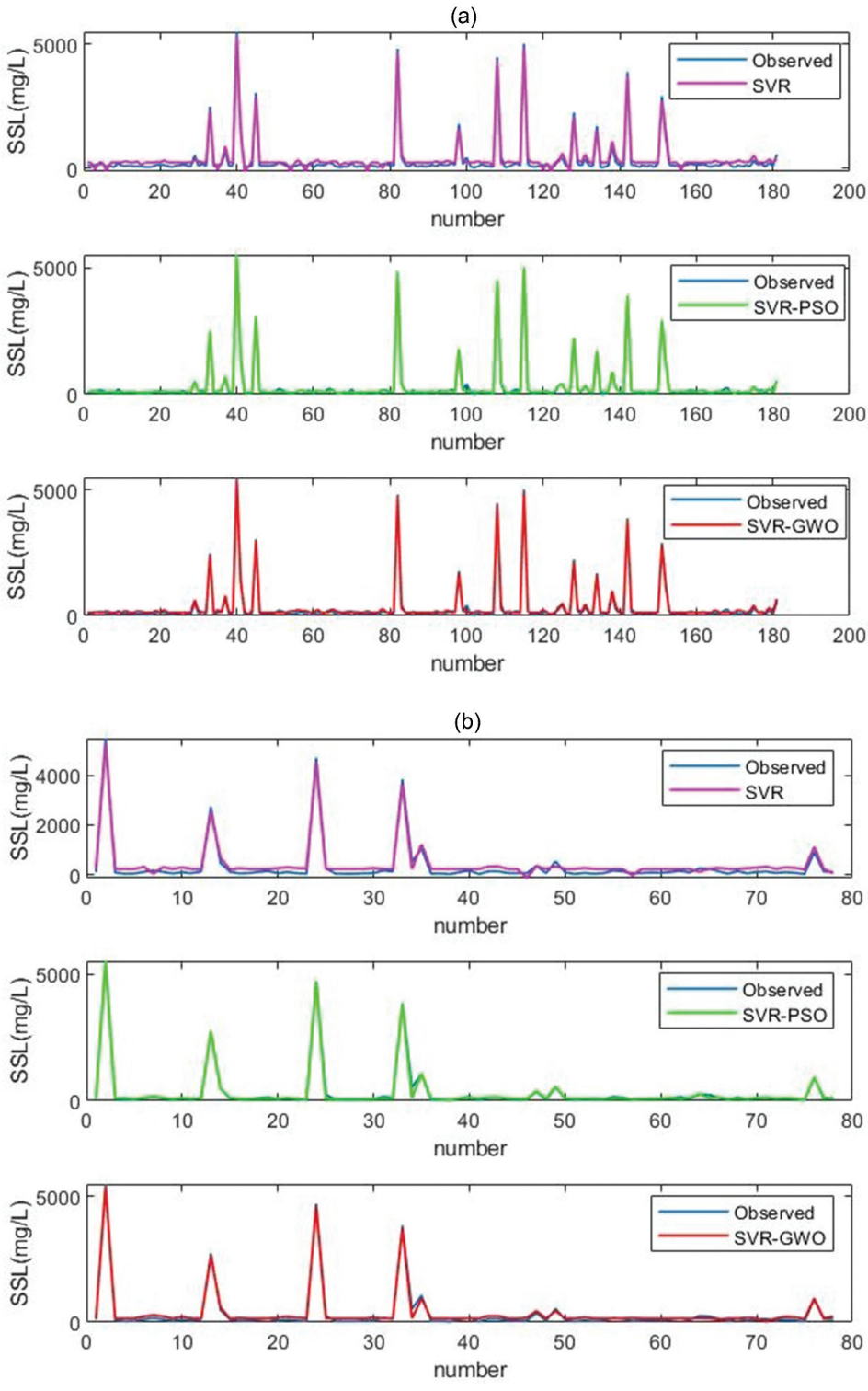


Figure 10. Actual daily suspended sediment load versus the estimates of different models: (a) training phase; (b) testing phase.

error predictability, and produces more appropriate performance of the predicted models.

Positive PBIAS values suggest the overestimation of the data (Moriasi *et al.* 2007). Therefore, all values are overestimated according to the data given in Tables 4–6. This issue can be solved by sediment classification. Despite the high difference between SRC and predictive models in all other criteria, it is superior to the two SVR-M7 and SVR-GWO-M7 models from the point of view of the PBIAS criterion, which may be due to the classification of sediment data in the SRC method.

Based on Fig. 10 and Table 7, the prediction is appropriately conducted using satellite data. However, with light sediment loads the model overestimates, so that the relative difference

Table 7. The quartiles values of the observed and estimated data.

Training phase	Observed SSL	SVR	SVR-PSO	SVR-GWO
$Q_{25\%}$	66.67	157.43	81.63	127.90
$Q_{50\%}$	99.33	159.48	101.74	128.34
$Q_{75\%}$	156.42	159.48	157.38	186.34
Testing phase	Observed SSL	SVR	SVR-PSO	SVR-GWO
$Q_{25\%}$	56.67	194.93	77.34	150.50
$Q_{50\%}$	97.75	199.03	101.04	151.04
$Q_{75\%}$	140.00	228.79	137.16	186.25

Table 8. Parameters of the sediment suspended load versus discharge regression relationship.

Type	Range (mg/L)	Parameters	
		a	b
I	3000<SSL≤6000	4239	0.01431
II	1000<SSL≤3000	802.8	0.2743
III	150<SSL≤1000	87.06	0.4069
IV	80<SSL≤150	86.43	0.04507
V	0<SSL≤80	65.51	-0.02575

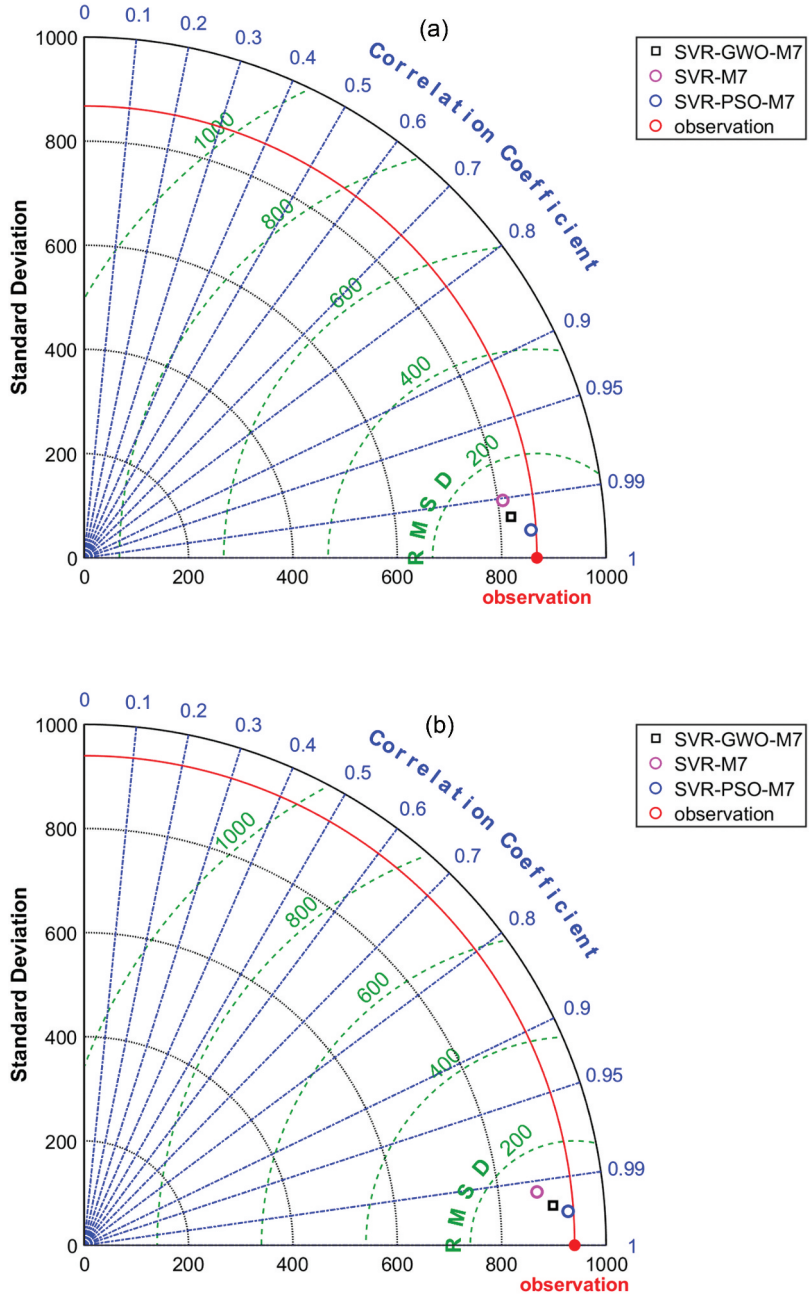


Figure 11. Taylor diagram of the observed and the predicted SSL (mg/L) for the SVR, SVR- PSO, and SVR-GWO models in the (a) training phase and (b) testing phase.

between the first quarter of the observed data and the estimated in the testing phase is 36.5%. The main reason for this difference is the lack of sediment data in the watershed. However, the model estimate is appropriate in 50 and 75% of the data and the heavy sediments, respectively. The results of this study are consistent with those of Himanshu *et al.* (2017). In the Himanshu study, only satellite precipitation products were used, and the best value was CC = 0.92. The superiority of

the results of the present research is likely due to the simultaneous use of the satellite precipitation and SM products. Unlike precipitation, SM products have no gaps and no zero values, making forecasting models better tuned. Therefore, satellite SM products have an influential role in SSL prediction.

7 Conclusion

This study evaluates the latest CHIRPS satellite precipitation and GLDAS SM products to estimate SSL. The study area was the Kosar Dam watershed in southwestern Iran. Information on this watershed is scarce, so sediment prediction is more difficult. To predict SSL, this study developed two combined models, SVR-PSO and SVR-GWO. The watershed was networked based on the spatial resolution of the precipitation products and the satellite SM. PCA was used to preprocess and reduce the size of the input data. Ten main components were selected as the input to the model. Ten prediction models each were designed for SVR, SVR-PSO, and SVR-GWO models. R, PBIAS, and RRMSE criteria are the main indicators used to determine the performance of the models. The combination of seven main components (M7) was selected as the best combination of inputs in all three methods. SSL was then estimated using a sediment rating curve. The results indicated the more appropriate performance of the M7 selected combination compared to the sediment rating curve method.

In summary, the following findings were obtained in this study:

(i) A good correlation was obtained between the estimated data and the observed dataset, such that in the SVR, SVR-GWO, and SVR-PSO prediction models, the CC values are higher than 0.99.
(ii) The predicted data of the hybrid models are closer to the observed data than the SVR model's output data.
(iii) Among the designed models, SVR-PSO-M7 has the highest performance in the testing and training stages.
(iv) The satellite precipitation and SM products are the main driving factors in watershed sediment prediction with limited information.

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